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An Approach for Detection and Localization of Missing Gate Faults in Reversible Circuit

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ABSTRACT

Power consumption issues in VLSI circuits resulted in the emergence of promising technologies – like reversible computing. The introduction of such computing paradigm has turned out to be beneficial for the application of low power CMOS design, communications and in the emerging area of quantum computing. With these benefits, physical realization of reversible circuits has received significant attention in the semiconductor industry. Additionally, to make a robust implementation of this technology, testing has been considered an essential component and several works on this filed are being reported on a daily basis. Although, many works on the efficient detection of gate faults exist but our main focus is not only to detect the faults in the circuit but also to localize the fault with 100% accuracy. Considering this aspect, in this article we propose a fault detection and localization scheme for the existing missing gate fault models in reversible circuits. In our approach, initially test vectors (*T*) are computed and then an algorithm executes to generate a unique test set (*U*). Once the test set (*U*) is generated, it is applied to detect any type of faults if exist in the circuit. Our approach has been successfully tested for all types of missing gate faults – *SMGF*, *PMGF* and *MMGF*. In way to verify the functional correctness of our scheme, we also have executed the testing policy over a wide spectrum of benchmarks and experimental findings are cross checked. The obtained results have been compared with peer-reviewed works and improvements are reported.

KEYWORDS

Missing-gate faults (MGF); Exact localization; Reversible circuit; Fault detection; Unique test set (U)

1. INTRODUCTION

According to Landauer's findings [1], traditional computing dissipates energy in form of heat for every information bit that is lost. Such information loss becomes a critical issue if the proposition of Moore's law [2] is preserved. Bennet [3] claims that no such energy will be dissipated if no information is lost while performing computation in the circuit. To ensure lossless computational feature, the circuit needs to satisfy the reversibility condition which can be achieved by using only reversible gates like Toffoli [4], Feynman [5] and Fredkin [6] gates and so on. Reversible logic not only resolves the power dissipation issues but also forms the basis for the emerging computing paradigm of quantum computing.

In respect to this, the synthesis of reversible circuits has received considerable attention of the researchers. For the practical realization of these circuits, all the constraints of reversibility have been considered and incorporated into the design algorithms. For instance, fan-out and feedback paths are strictly prohibited. Therefore, design of reversible circuits differs from the traditional one and thereby various design solutions have been introduced [7–9]. Along with this, prevention and detection of faults in the successful implementation of these circuits have also become significant. Especially, this turns out to be critical for quantum computation as the states of the qubit can be easily collapsed due to quantum mechanical property of de-coherence, which ultimately leads to the loss of information and thereby causes faults.

Consequently, distinct fault models with their corresponding test generation algorithms have been investigated in [10–17]. The fault models [18–20] which have been extensively studied for investigating defects in quantum *k*-CNOT gates includes single missing fault (*SMGF*), repeated gate fault (*RGF*), partial missing gate fault (PMGF) and multiple missing gate fault (*MMGF*). These models provide a better realization of all the possible types of failures that can occur in a reversible circuit model. In this article, we have introduced a fault detection approach for the existing fault models and

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also have determined the location of the corresponding fault.

2. PRELIMINARIES

Here we introduce the fundamentals on reversible logic like reversible functionality, gates, circuits and different fault models.

2.1 Reversible Function

Definition 1: *A function having the identical number of inputs and outputs, where the generated vectors satisfy the bijective mapping condition is termed as reversible functions.*

In reversible circuit no fan-out or feedbacks are allowed and the circuit can be operated from both the directions.

2.2 Reversible Gates

Definition 2: *Reversible gates are considered as the basic component of reversible circuit and operate on the input vectors to produce the corresponding output vectors based on the provided specification function.*

The different types of elementary gates used for the synthesis of reversible circuits are listed below.

NOT gate *– A single input/output gate that inverts the input logic value from 1 to 0 and 0 to 1. It can be realized as* $(X_1 \rightarrow \bar{X}_1)$.

CNOT gate *– This gate has one control and a single target node. This target node inverts the input bit once the control is set to 1. The gate can be expressed in the following vector form* $(X_1, X_2) \rightarrow (X_1, X_2 \oplus X_1)$.

Toffoli gate *– It is a three input/output gate that performs the inverse operation in target line only when both of its control inputs are assigned to logic 1. This gate can be expressed in the following form* $(X_1, X_2, X_3) \rightarrow (X_1, X_2, X_3 \oplus X_1.X_2)$.

Definition 3: *A generalized k-CNOT gate has a set of k control inputs* X_1, X_2, X_3, $\ldots$,X_k *and only a single target input t. The corresponding transformation function from input to output can be represented as* $(X_1, X_2, X_3, \ldots, X_k, t) \rightarrow (X_1, X_2, X_3, \ldots, X_k, t \oplus X_1.X_2.X_3 \ldots X_k)$. *The k number of control inputs pass through the gate without any change in bit value where the target input t is inverted and becomes the (k+1)th output once all the k controls are set at logic 1.*

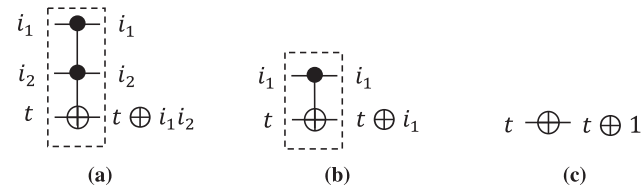


Figure 1: (a) 2-control Toffoli gate (b). *CNOT* gate (c) *NOT* gate

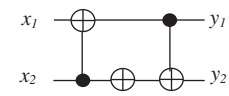


Figure 2: Example of reversible circuit

Table 1: Reversible specification of Figure 2

INPUTS (x_1x_2)	OUTPUTS (y_1y_2)
0	$f(0) = 1$
1	$f(1) = 3$
2	$f(2) = 2$
3	$f(3) = 0$

The schematic representations of the previously mentioned gates are shown in Figure 1.

2.3 Reversible Specification of Circuit

The reversible specification of a circuit is an input-output combination in which the output is the permutation of the input. A circuit (RS_{CKT}) having n input lines and N number of gates can have 0 to (2^n-1) different input combinations for its n input lines. Hence, the reversible specification of the circuit would be as follows:

$$RS_{CKT} = \begin{pmatrix} 0 & 1 & \ldots & 2^n - 1 \\ f(0) & f(1) & \ldots & f(2^n - 1) \end{pmatrix}$$

where $f(i)$ is the function that computes the output value for the corresponding input value i, where $0 \leq i \leq 2^n-1$.

Example 1: Let us consider the circuit depicted in Figure 2 that explains the behavior of reversible specification.

For the input value $i = 0$, the circuit generates output as 1 *i.e.* $f(0) = 1$. Similarly, other values are computed as $f(1) = 3, f(2) = 1, f(3) = 1$. The input-output combination (decimal form) of the given circuit has been represented in Table 1.

Hence, the reversible specification (RS_{CKT}) of the given circuit is as follows:

$$RS_{CKT} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ f(0) & f(1) & f(2) & f(3) \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 \end{pmatrix}$$

2.4 Fault Models of Reversible Circuit

Here, we introduce the major gate fault models (SMGF, PMGF, and MMGF) that may exist in reversible circuit.

2.4.1 Single missing gate fault (SMGF)

Definition 4: *This type of fault occurs when a single k-CNOT gate disappears from the circuit by changing the circuit's original functionality.*

If we consider the circuit of Figure 3(a) assuming that the gate marked within a dotted box in that figure is missing from the given circuit, then the input vectors (for example ⟨0, 1, 1⟩) which assign a logic value 1 at the control inputs of that gate detect the SMGF from the circuit. The possible number of SMGF faults in a circuit is equivalent to total gate count of the circuit.

2.4.2 Repeated gate fault (RGF)

Definition 5: *This fault defines repeating a gate any number of times in a circuit. An even number of occurrences of a gate makes an RGF fault equivalent to that of an SMGF fault but an odd number of occurrences make the circuit faulty and the faulty largely remain undetected.*

A circuit containing RGF fault is shown in Figure 3(b) where the first gate is replaced and repeated twice in the given circuit. The detection condition of this fault model is similar to that of an *SMGF* in which applying only logic 1 at the control lines confirms the presence of the fault.

2.4.3 Multiple missing gate fault (MMGF)

Definition 6: *If multiple gates go missing at consecutive positions from a circuit then the induced fault is termed as MMGF.*

An example on MMGF is shown in Figure 3(c), where the first two gates (enclosed within the box) disappear from the circuit by causing MMGF in the design. Again, once the test vector ⟨0, 1, 1⟩ is applied at the input of the faulty circuit, the presence of the fault is confirmed. But a complete execution of SMGF test does not ensure the detection of all possible occurrences of MMGF faults in the circuit.

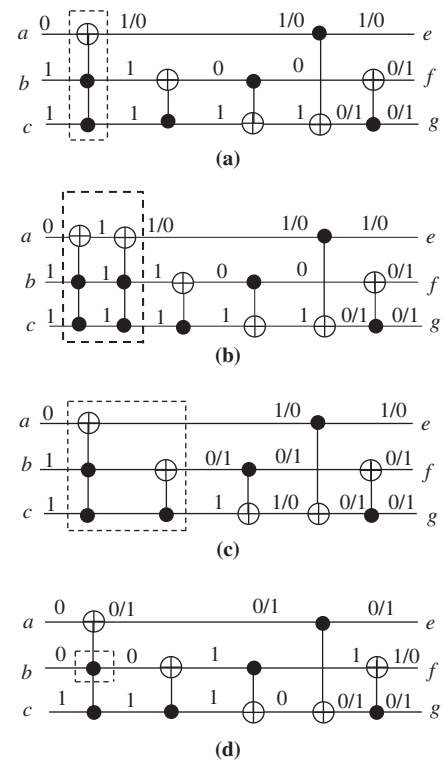


Figure 3: (a) Testing for possible *SMGF* in *ham3ndesign#1* using test vector ⟨0, 1, 1⟩ (b) Testing for possible RGF in *ham3ndesign#1* using test vector ⟨0, 1, 1⟩ (c) Testing for possible MMGF in *ham3ndesign#1* using test vector ⟨0, 1, 1⟩ (d). Testing for possible PMGF in *ham3ndesign#1* using test vector ⟨0, 0, 1⟩

2.4.4 Partial missing gate fault (PMGF)

Definition 7: *In-spite of missing the complete gate from the circuit, if some control nodes go missing from a certain number of gates then the incurred fault is known as PMGF. The number of missing controls denotes the order of the PMGF.*

Under this fault the functionality of the circuit changes completely as if we assume that p number of control misses from a k-CNOT gate then it changes the gate specification to $(k\text{-}p)$-CNOT gate which seriously affect the overall functionality of the circuit.

Consider the circuit of Figure 3(d), where a first order *PMGF* occurs due to the absence of a control input at the first gate (whose position has been marked with a box). Once we pass such a test vector (let it be ⟨0, 0, 1⟩) that set logic value 0 at the missing control positions then the fault is identified. But it is important to remember that for the detection of all possible k-number of first order faults, a set of k test vectors are required. Furthermore, these test vectors also can ensure detection of all the possible higher order faults as well.

3. RELATED WORK AND CONTRIBUTION

We have already discussed about the different types of fault models that can occur in a reversible circuit and the corresponding test vectors needed for their identification. Direct application of these test vectors does not seem to be the best way to detect faults and thereby several works with different testing strategies have been reported. Basically, much of these testing works are using test pattern generators while some others are based on testable design models. Here, we are reviewing some of those testing schemes.

Fault detection using deterministic and probabilistic tests in quantum reversible circuits has been discussed in [19]. As the testing process involves extensive computations, so to reduce this overhead in fault analysis, in [21], an integer linear programming (ILP) based minimal test set generation technique for multiple stuck-at fault model has been shown and it is found very effective in fault diagnosis. Furthermore, to minimize the test vectors requirement, in [16], an approach based on optimal test set (OTS) has been introduced.

However, an uncertainty always remains about these test algorithms regarding the attainment of full (100%) fault coverage and to ensure this, a new testing approach namely "Ping Pong test" has been introduced in [22]. An efficient way of fault detection and localization in reversible circuits based on adaptive tree-based model has been introduced in [10].

To make the test vector generation method simpler, an automatic generation of test patterns for the identification of various missing gate faults has been discussed in [20]. Detection of faults in quantum k-CNOT gates using copy gate method and by embedding additional ancilla lines is developed in [23]. To improve the test generation procedure further, a new testing approach based on Boolean difference technique is shown in [13].

Detection and identification of SMGFs in ESOP based reversible circuit is described in [24], where the Reed-Muller form is applied to test the fault in the circuit.

Without taking out the circuit from operational mode, an online testing approach is introduced in [25], where two 4×4 reversible gates are used to find out single bit faults in reversible circuits.

Extending this work further, in [26], a parity bit based testable block (TB) is implemented for efficient localization of faults in online mode.

By introducing the concept of universal reversible gate (URG), in [27], an improved online testing scheme is developed and this method not only checks gate faults but efficiently catches errors like single-bit error, soft error *etc*.

All the testing strategies discussed so far are based only on fault detection in reversible circuits. But, the localization of these detected faults still remains an important issue which needs to be addressed. For this purpose, algorithms are required to localize the faults so that a better diagnosis of the faulty circuit can be performed.

Motivated by this idea, here, in this work, we have developed an algorithm which not only determines the fault effectively but also locate the faculty area with 100% accuracy. In our testing process, initially, test vectors are generated by comparing the reversible specification of the fault free and faulty circuit and in later stage, using Boolean difference method unique test vectors are derived for each possible missing gate in the circuit. Next, all these unique vectors are used for both detection and localization of the faults in the circuit. In this work, we have covered all type of major gate faults *i.e. SMGF*, *PMGF*, *MMGF* and our testing method has been experimentally verified over a wide range of benchmarks to justify the effectiveness of our scheme.

The rest of the work is formulated as follows. The detailed overview of our proposed testing approach has been explained in Section 4. The computation complexity of our approach is derived in Section 5. The experimental results with analysis are summarized in Section 6 and finally the concluding remarks appear in Section 7.

4. PROPOSED FAULT DETECTION AND LOCALIZATION MODEL

In this section, fault detection and localization process for major gate faults are discussed in detail. Each of the testing schemes is completed in two phases – first, we have computed the Boolean expression for the fault free circuit and then have executed the fault specific diagnosis algorithms.

4.1 Generation of Boolean Expression for the Fault Free Reversible Circuit

Sequences of reversible gates are used in a reversible circuit to realize any specific reversible function. The number of such gates in a circuit determines the corresponding circuit depth which in turn depends on the number of levels exist. For instance, a circuit with N gates

has N levels starting from 0 to $(N-1)$. Each of these levels contains a reversible gate where a gate at the ith level is denoted as G_i for i in $0 \leq i \leq (N-1)$. Every input line *(I)* of the circuit is examined from the left to find out the target nodes. When a target node (t_g) of any gate g is found, then the control node position (c_g) of the corresponding gate is determined and used to compute the output Boolean expression of that gate. This process is repeated for all the gates present over the input line I and the final output expression for I corresponds to the Boolean expression of the last target node appearing over the same line I.

To get a better apprehension of the above said method, next we have explained it with an example.

Example 2: Let us consider the circuit shown in Figure 4 which has 3 input (x_1, x_2 and x_3) lines and 2 gates (G_0 and G_1) placed at *level* 0 and *level* 1 over the three circuit lines. Gate G_0 has target node at line x_1 and control at x_3. Similarly, gate G_1 has target node at x_3 and control nodes at x_1 and x_2.

The final output expression at y_1 is the output expression from the target node of that gate at *level* 0. The output at y_2 is equal with the input at port x_2 as there is no target node available on line x_2. Similarly, the final output expression at y_3 is the output expression of the target node of G_1 at *level* 1 of that circuit. Now, in Table 2 we summarize the output expressions for each line.

4.2 Algorithm for Detection and Localization of SMGF

Here, we discuss the fault localization policy for *SMGF* in reversible circuit. Before elaborating our testing model, here in Table 3, we introduce some interpretations and basic notations related to our testing algorithm.

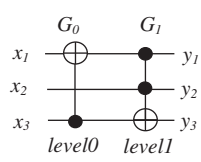


Figure 4: Example circuit

Table 2: Computed Boolean expression for Figure 4

Line Number	Corresponding Boolean Expression
0	$y_1 = (x_1 \oplus x_3)$
1	$y_2 = x_2$
2	$y_3 = ((x_1 \oplus x_3).x_2) \oplus x_3$

Table 3: List of notations and their interpretations

Notations	Definition
C_{TEST}	Testable circuit (may be fault-free or faulty)
C_{FF}	Fault –free circuit of C_{TEST} circuit
RS_{FF}	Input-output combination of the C_{FF} circuit
$RS_{FF}(K)$	Input-output combination of the C_{FF} circuit for the input value K
$P_{TEST}(K)$	Input-output combination of the C_{TEST} circuit for the input value K
T_{G_i}	Test set for the detection of each of the possible missing gate G_i at i^{th} level of the C_{FF} circuit
O_j	Number of times the vector J occurs in different T_{G_i} is called Occurrence Factor of J
U_{G_i}	Unique test set for the detection of each possible missing gate G_i at i^{th} level of the C_{FF} circuit
T_L	It is the test set which is used to locate the fault in the circuit

Let consider a circuit containing N number of gates over n input lines, where the symbol T, O, U used in the table represents test vector, occurrence factor and unique test set for a gate G_i for $0 \leq i \leq (N-1)$.

Now, here, we state all the involved phases one by one.

Step 1: In the very first step, the reversible specification (RS_{FF}) of the fault free circuit (C_{FF}) is computed.

Step 2: The (2^n-1) possible inputs are applied to the C_{FF} and test set (T_{G_i}) for each of the possible missing gates (G_i) of C_{FF} is generated by setting "1" at all the control nodes and either "0" or "1" to the target node and through line of the G_i respectively, where $0 \leq i \leq (N-1)$ and N is the number of gates in the C_{FF} over n input lines.

Step 3: The complete test set *(T)* for detecting all possible *SMGF* of C_{FF} is determined by taking the union of all T_{G_i} for each G_i obtained in step2.

So, $T = T_{G_0} \cup T_{G_1} \cup T_{G_2} \cup \ldots \cup T_{G_{N-1}}$.

Step 4: The occurrence factor (O_J) for each vector J in T is obtained by searching each T_{G_i} from the circuit.

Step 5: The occurrence factor (O_J) obtained in step4 is stored in a table (known as occurrence factor table) to check the occurrence of different values in T.

Step 6: Now, the unique test set (U_{G_i}) for each gate G_i is formed using T_{G_i}. The value for U_{G_i} is obtained by combining minimum occurrence test vectors with another equal or higher occurrence test vectors in T_{G_i}. Here, m number of test vectors from T_{G_i} need to be combined to form U_{G_i}, where m is the minimum occurrence factor of a test vector in T_{G_i}.

Step 7: The test set (T_L) for localization of the fault in the circuit is formed as follows:

$$T_L = U_{G_0} \cup U_{G_1} \cup U_{G_2} \cup \ldots \cup U_{G_{N-1}}.$$

Basically, this parameter T_L is the reduced form of factor T.

Step 8: Here detection and localization of any *SMGF* is to be determined on the testable circuit C_{TEST}. In this step, we apply a random test vector K from the vector list T_L over the circuit C_{TEST} and store the output in $P_{TEST}(K)$. If the condition $RS_{FF}(K) \neq P_{TEST}(K)$ holds then we conclude that a fault of type *SMGF* exists in the circuit. Further, to trace the corresponding location of *SMGF* in C_{TEST}, T_{G_i} is searched in such a way that $K \in T_{G_i}$.

All the above mentioned phases involved for detection and localization of SMGF in reversible circuit are summarized step by step in Algorithm 1.

Example 3: For easy of understanding, here we explain all the mentioned phases using an example circuit. Now, consider the circuit depicted in Figure 5(a) which contains five gates over three circuit lines {a, b, c}.

Step1: Here, first we compute the input-output combination (RS_{FF}) for the circuit C_{FF} and the computed sequence is:

Step 2: Now, by reading the circuit specification, we assign the values for different circuit parameters where we set $n=3$, $N=4$ and $i=4$. Once we pass the two test vectors ($\langle 0, 1, 1\rangle$ and $\langle 1, 1, 1\rangle$) out of eight patterns through the circuit C_{FF}, then all the control nodes of gate G_0 and target node of G_0 receive 0 and 1, respectively (see in Figure 5(c)), which in turn confirms the presence of SMGF due to missing of a gate G_0 from C_{FF}. So, we compute T_{G_0} as $T_{G_0} = \langle 3, 7\rangle$.

In the similar way, the test set for the remaining gates are obtained as $T_{G_1} = \langle 1, 3, 5, 7\rangle$, $T_{G_2} = \langle 1, 2, 5, 6\rangle$, $T_{G_3} = \langle 3, 4, 5, 6\rangle$, $T_{G_4} = \langle 2, 4, 5, 7\rangle$.

Step 3: Next, we compute the test set T for the whole circuit and it becomes $T = T_{G_0} \cup T_{G_1} \cup T_{G_2} \cup T_{G_3} \cup T_{G_4}$.

$$T = \langle 3,7\rangle \cup \langle 1,3,5,7\rangle \cup \langle 1,2,5,6\rangle \cup \langle 3,4,5,6\rangle \cup \langle 2,4,5,7\rangle$$

$$T = \langle 1, 2, 3, 4, 5, 6, 7\rangle.$$

Step 4: In this step, we compute the occurrence factor O_1 for $J = 1$ by searching it from the set $T_{G_0}, T_{G_1}, T_{G_2}, T_{G_3}$ and T_{G_4} to check the presence of value 1 contained in the set T.

It can be observed that the value 1 belongs to T_{G_1} and T_{G_2}. Hence, occurrence factor O_1 is 2 *i.e* $O_1 = 2$. In a similar manner, the occurrence factors for the remaining values in T are obtained and they are $O_2 = 2$, $O_3 = 3$, $O_4 = 2$, $O_5 = 4$, $O_6 = 2$ and $O_7 = 3$.

Step 5: Now, all the derived occurrence factors are populated in a list (here it is Table 4).

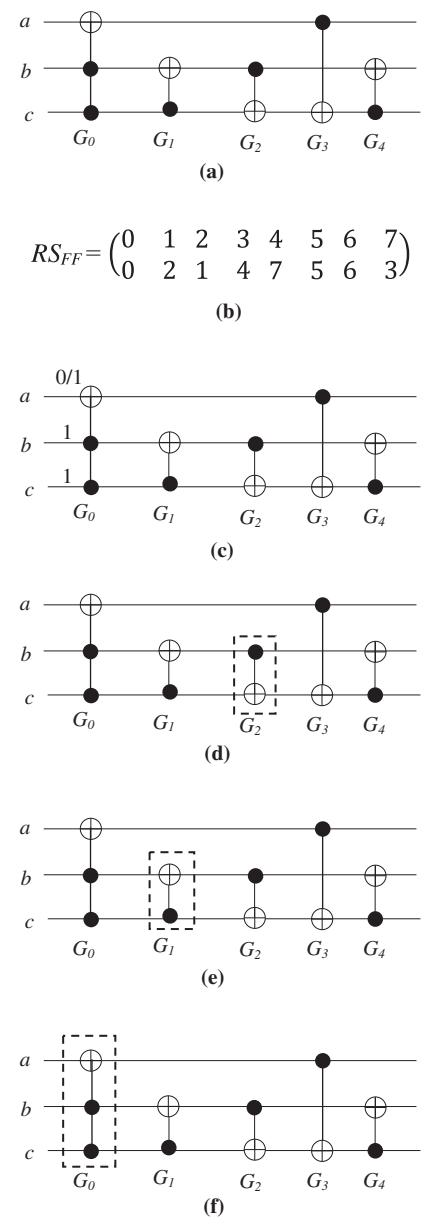


Figure 5: (a) Fault free circuit (C_{FF}) (b) Input-output combination of C_{FF} of Figure 5(a) (c) Test vector generation for G_0 of C_{FF} (d) Testable Circuit (C_{TEST}) for SMGF (e) Testable Circuit (C_{TEST}) for SMGF 5(f). Testable Circuit (C_{TEST}) for SMGF

Table 4: Occurrence Factor Table for *SMGF*

Test Values in T	Occurrence Factor (O_J)
1	2
2	2
3	3
4	2
5	4
6	2
7	3

Step 6: Here, the unique test sets are computed for each of the gate G_i present in the testable design. To compute U_{G_0}, $T_{G_0} = \langle 3, 7\rangle$ is used and the occurrence factor for each of the values (3, 7) inT_{G_0} is searched from Table 4. Now, it can be observed that both 3 and 7 have same occurrence factors of 3, *i.e.* $O_3 = O_7 = 3$. So, the value for m is 3 and we need to combine 3 vectors from T_{G_0} to form U_{G_0} but it is not possible as $T_{G_0} = \langle 3, 7\rangle$ is constructed with only two test vectors. If we take a look at T_{G_1}, T_{G_2}, T_{G_3} and T_{G_4} then we can see that $T_{G_0} \subseteq T_{G_1}$, hence in this situation we need to form U_{G_1} and then any one value from the U_{G_1} is to be combined with $T_{G_0} = \langle 3, 7\rangle$ to cross check the location of the fault. So, we need to focus on the formation of U_{G_1}.

In the similar way the unique test set for the remaining gates are obtained as follows:

$$U_{G_1} = \langle 1, 3\rangle,$$

$$U_{G_2} = \langle 1, 2\rangle,$$

$$U_{G_3} = \langle 4, 6\rangle,$$

$$U_{G_4} = \langle 2, 4\rangle.$$

Step 7: In this step, unique test vectors for entire circuit T_L is obtained as

$$T_L = U_{G_0} \cup U_{G_1} \cup U_{G_2} \cup U_{G_3} \cup U_{G_3}$$

$$T_L = \langle 3, 7\rangle \cup \langle 1, 3\rangle \cup \langle 1, 2\rangle \cup \langle 4, 6\rangle \cup \langle 2, 4\rangle$$

$$T_L = \langle 1, 2, 3, 4, 6, 7\rangle$$

Step 8: Again, if we assume that gate G_2 is missing (shown using a dotted box in Figure 5(d)) from the circuit, the steps as stated in 1–6 accurately detect the fault but cannot identify the location. Now, we will locate it next.

ALGORITHM 1: Detection and Localization of SMGF in Reversible Circuit

Input: A testable reversible circuit (C_{TEST}).
Output: Exact level where fault is present or name of the gate in which fault is present.
begin
Step1: **Computation of Reversible Specification (RS_{FF}):** Reversible specification (RS_{FF}) of the fault free circuit (C_{FF}) is computed.
Step2: **Test Vector (T_{G_i}) Generation for Each Gate (G_i):** (2^n-1) possible inputs are applied to the C_{FF} and test set (T_{G_i}) for each possible missing gate (G_i) of C_{FF} is generated by setting '1' at all the control nodes and either '0'or '1' to target node and through line of the G_i, where $0 \le i \le (N\text{-}1)$.
Step3: **Complete Test Set (T) for All Possible *SMGF*:** Complete test set (T) of the circuit as follows:
$T = T_{G_0} \cup T_{G_1} \cup T_{G_2} \cup \ldots \cup T_{G_{N-1}}$.
Step4: **Computation of Occurrence Factor (O_J):** Occurrence factor (O_J) of each vector (J) in T is calculated from different T_{G_i}, where $0 \le i \le (N\text{-}1)$ and $J \in T$.
Step5: **Formation of Occurrence Factor Table:** The occurrence factor table stores occurrence of different vector of T in different T_{G_i}, where $0 \le i \le (N\text{-}1)$.
Step6: **Computation of Unique Test Set (U_{G_i}) for Each Gate:** Compute unique test vector (U_{G_i}) for each gate G_i by combining lowest occurrence test vector with another equal or higher occurrence test vector in T_{G_i}.
Step7: **Computation of Test Set for Fault Localization (T_L):** Test set (T_L) of the circuit is computed as follows:
$T_L = U_{G_0} \cup U_{G_1} \cup U_{G_2} \cup \ldots \cup U_{G_{N-1}}$.
Step8: ***SMGF* Detection and Exact Localization:** Apply any test vectors (K) randomly from $T_L$on the C_{TEST} and store the output in $P_{TEST}(K)$, where $K \in T_L$.
IF ($RS_{FF}(K) \neq P_{TEST}(K)$) THEN
begin
8.1 *SMGF* fault is present in C_{TEST}.
//Detection of SMGF in C_{TEST}//
8.2 T_{G_i} is searched such that $K \in T_{G_i}$ and corresponding U_{G_j} of each T_{G_i} would be used to localize the fault.
//Exact Localization of *SMGF* in C_{TEST}//

Here, first we select a random value (K) from the set $T_L = \langle 1, 2, 3, 4, 6, 7\rangle$ and let it be 4. Now, this value is applied to the circuit C_{TEST} and it is observed that, $RS_{FF}(4) = P_{TEST}(4)$, so it is confirmed that the fault cannot be detected using the random value 4. Next, we take another random value 6 for K and find that $RS_{FF}(6) \neq P_{TEST}(6)$, which further confirms the presence of *SMGF* in C_{TEST}. Now, our aim is to locate the gate which is missing from the C_{TEST}.

To find out the exact location of the fault in C_{TEST}, initially we need to find out T_{G_i} in which $K = 6$ is present. It can be seen from phase 2 that $K = 6$ belongs T_{G_2} and T_{G_3}. So, the fault is present either in G_2 or G_3 and the set $U_{G_2} = \langle 1, 2\rangle$ and $U_{G_3} = \langle 4, 6\rangle$ are required now to find the location of the fault. We already have seen that $K = 4$ could not detect the fault and this value 4

Table 5: Notations and their interpretations for Algorithm 2

Notations	Definition
(T_{G_i}, x)	Test set for the detection of possible missing controls at input line x of the gate G_i at i^{th} level of the C_{FF} circuit.
(U_{G_i}, x)	Unique test set for the detection of possible missing controls at input line x of gate G_i at i^{th} level of the C_{FF} circuit.

exists in $U_{G_3} = \langle 4, 6\rangle$, so it is confirmed now that the fault is not present in G_3. Therefore, it can be concluded that *SMGF* has been caused due to the missing of gate G_2.

In the similar way, the missing of gate G_1 from Figure 5(e) can be located by setting K value to 1, but if K becomes 3 then finding the faulty gate is not possible.

Using the same technique, the missing gate G_0 from Figure 5(f) is tracked using $K = 1$.

4.3 Algorithm for Detection and Localization of PMGF

Like we have done for SMGF in Algorithm1, here, we dealt with the identification and localization of PMGF type faults.

The basic notations and terms used in our technique are summarized in Table 5.

For easy of understanding, first the entire PMGF detection and localization scheme is explained using an example and then we have summarized all the involved steps sequentially in Algorithm2.

Example 4: Now again let consider the same circuit (C_{FF}) of Figure 5(a) as an example benchmark.

Step 1: Under this phase, like we have done for SMGF, here first we derive the input-output combinations (RS_{FF}) for the circuit C_{FF}.

Step 2: The possible patterns that can be tested over C_{FF} range between 0 and 7, but as per the detection condition of *PMGF*, we only can apply input vector $\langle 0, 0, 1\rangle$ and $\langle 1, 0, 1\rangle$ to detect possible missing controls at line b of G_0 in circuit C_{FF}, so, $(T_{G_0}, b) = \langle 1, 5\rangle$.

Similarly, the other test sets are computed as $(T_{G_0}, c) = \langle 2, 6\rangle$, $(T_{G_1}, c) = \langle 0, 2, 4, 6\rangle$, $(T_{G_2}, b) = \langle 0, 3, 4, 7\rangle$, $(T_{G_3}, a) = \langle 0, 1, 2, 7\rangle$ and $(T_{G_4}, c) = \langle 0, 1, 3, 6\rangle$.

Table 6: Occurrence Factor Table for PMG*F*

Test Values in T	Occurrence Factor (O_J)
0	4
1	3
2	3
3	2
4	2
5	1
6	3
7	2

Step 3: In this phase, the test set (T) for the whole circuit is derived as -

$$T = (T_{G_0}, b) \cup (T_{G_0}, c) \cup (T_{G_1}, c) \cup (T_{G_2}, b) \cup (T_{G_3}, a) \cup (T_{G_4}, c)$$

$$T = \langle 1, 5\rangle \cup \langle 2, 6\rangle \cup \langle 0, 2, 4, 6\rangle \cup \langle 0, 3, 4, 7\rangle \cup \langle 0, 1, 2, 7\rangle \cup \langle 0, 1, 3, 6\rangle$$

$$T = \langle 0, 1, 2, 3, 4, 5, 6, 7\rangle.$$

Step 4: Now, we compute the occurrence factor of O_0 for $J = 0$ and search the entire spectrum $\{(T_{G_0}, b), (T_{G_0}, c), (T_{G_1}, c), (T_{G_2}, b), (T_{G_3}, a), (T_{G_4}, c)\}$ and find that 0 belongs to set (T_{G_1}, c), (T_{G_2}, b), (T_{G_3}, a) and (T_{G_4}, c). Therefore, O_0 is 4 *i.e* $O_0 = 4$.

Similarly, rest of the occurrence factors are computed and they are:

$$O_1 = 3, O_2 = 3, O_3 = 2,$$
$$O_4 = 2, O_5 = 1, O_6 = 3,$$
$$O_7 = 2.$$

Step 5: Under this action, we append all the occurrence factors for T in a separate list and here, the list is presented in Table 6.

Step 6: Now we require to form (U_{G_0}, b). To compute the set, first, we select the rudimentary value for (T_{G_0}, b) as $\langle 1, 5\rangle$ but it is observed from Table 6 that $O_1 = 3$ and $O_5 = 1$. Interestingly, the value for $m = O_5 = 1$, so only a single value (5) is required to form the (U_{G_0}, b) from (T_{G_0}, b). Hence, $(U_{G_0}, b) = \langle 5\rangle$.

In the similar way, when we try to compute the value for (U_{G_0}, c) from $(T_{G_0}, c) = \langle 2, 6\rangle$, then a problem occurs as (T_{G_0}, c) is contained within $(T_{G_1}, c) = \langle 0, 2, 4, 6\rangle$, so, the computation process for (U_{G_0}, c) needs to be postponed.

By following the previously stated strategy, we find the values for rest of the gates which are listed below.

ALGORITHM 2: Detection and Localization of *PMGF* in Reversible Circuit

Input: A testable reversible circuit (C_{TEST}).
Output: Exact level where fault is present or name of the control node where fault is present.

begin

Step1: **Computation of Reversible Specification (RS_{FF}):** Reversible specification (RS_{FF}) of the fault free circuit (C_{FF}) is computed.

Step2: **Test Vector (T_{G_i}, a) Generation for Each Control of The Gate (G_i):** (2^n-1) possible inputs are applied to the C_{FF} and test set (T_{G_i}, x) for each possible missing of control node of gate (G_i) at the input line a of C_{FF} is generated by setting '0' at the possible missing control node and '1' at the all other control nodes and either '0' or '1' to target node and through line of the G_i, where $0 \le i \le (N-1)$ and x refers to input lines of the circuit.

Step3: **Complete Test Set (T) for All Possible *PMGF*:** Complete test set (T) of the circuit as follows: $T = (T_{G_0}, a) \cup (T_{G_0}, b) \cup (T_{G_1}, a) \cup (T_{G_1}, b) \cup \ldots \cup (T_{G_{N-1}}, a) \cup (T_{G_1}, b)$, where a, b are control nodes of different gates.

Step4: **Computation of Occurrence Factor (O_J):** Occurrence factor (O_J) of each vector (J) in T is calculated, where $J \in T$.

Step5: **Formation of Occurrence Factor Table:** The occurrence factor table stores occurrence of different vector of T.

Step6: **Compute Unique Test Set (U_{G_i}, x) for Each Fault:** Compute unique test vector (U_{G_i}, x) for possible missing of control node at line x of gate G_i by combining lowest occurrence test vector with another equal or higher occurrence test vector in (T_{G_i}, x), where x refers to input lines of the circuit.

Step7: **Computation of Test Set for Fault Localization (T_L):** Test set (T_L) of the circuit is computed as follows: $T_L = (U_{G_0}, a) \cup (U_{G_0}, b) \cup (U_{G_1}, a) \cup (U_{G_1}, b) \cup \ldots \cup (U_{G_{N-1}}, a) \cup (U_{G_1}, b)$.

Step8: ***PMGF* Detection and Exact Localization:** Apply any test vectors (K) randomly from T_L on the C_{TEST} and store the output in $P_{TEST}(K)$, where $K \in T_L$.

IF ($RS_{FF}(K) \neq P_{TEST}(K)$) THEN

begin

8.1 *PMGF* fault is present in C_{TEST}.
//Detection of PMGF in C_{TEST}//

8.2 (T_{G_i}, x) is searched such that $K \in (T_{G_i}, x)$ and corresponding (U_{G_j}, x) of each (T_{G_i}, x) would be used to localize the fault, where x refers to input lines of the circuit.
//Exact Localization of *PMGF* in C_{TEST}//

$$(U_{G_1}, c) = \langle 4, 6\rangle, (U_{G_2}, b) = \langle 3, 4\rangle,$$
$$(U_{G_3}, a) = \langle 1, 7\rangle, (U_{G_4}, c) = \langle 3, 6\rangle.$$

But the test set (U_{G_0}, c) is yet to be obtained and as the vector (T_{G_0}, c) = ⟨2, 6⟩ is the subset of (T_{G_1}, c) = ⟨0, 2, 4, 6⟩, so cross-checking needs to be done with vector 4 to confirm the disappearance of control node from line c of gate G_0 and finally, it is found that (U_{G_0}, c) = ⟨2, 6⟩.

$$RS_{FF} = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 2 & 1 & 4 & 7 & 5 & 6 & 3 \end{pmatrix}$$

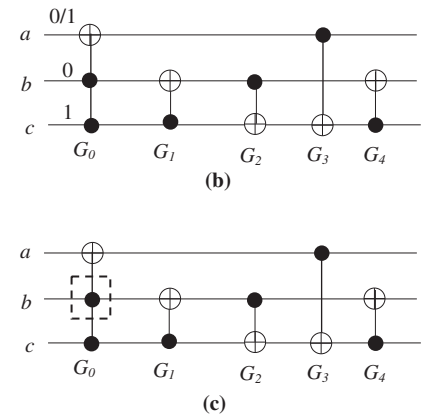


Figure 6: (a). Input-output combinations for C_{FF} of Figure 5(a) (b) Test vector generation for the control at line b of G_0 in C_{FF}(c) Testable Circuit (C_{TEST}) for PMGF

Step 7: Now, we compute the test set T_L for the entire circuit as

$$T_L = (T_{G_0}, b) \cup (T_{G_0}, c) \cup (T_{G_1}, c) \cup (T_{G_2}, b) \cup (T_{G_3}, a) \cup (T_{G_4}, c)$$
$$T_L = \langle 5\rangle \cup \langle 2, 6\rangle \cup \langle 4, 6\rangle \cup \langle 3, 4\rangle \cup \langle 1, 7\rangle \cup \langle 3, 6\rangle$$
$$T_L = \langle 1, 2, 3, 4, 6, 7\rangle$$

Phase 8: Next, we localize the fault in the circuit of Figure 6(c). Let's consider that the control node at line b of gate G_o is missing (shown as dotted box in Figure 6(c)) from the circuit.

So, first we randomly choose a value for K (let it be 5) from the set $T_L = (1, 2, 3, 4, 5, 6, 7)$ and apply it to C_{TEST} of Figure 6(c), where we find that for $K = 5$, $RS_{FF}(5) \neq P_{TEST}(5)$ and this condition confirms the presence of *PMGF* in C_{TEST}. While locating the fault in the circuit again when we look back to the step2 of the scheme, it is seen that the $K = 5$ belongs to sets (T_{G_0}, b) and (U_{G_0}, b). So, it confirms that the fault has occurred at line b of G_0.

So, we have successfully detected and localized the PMGF in the circuit. By following the same procedure, the different first order PMGFs in rest of the gates can be tracked too.

4.4 Algorithm for Detection and Localization of MMGF

The scheme for exact localization and detection of MMGF is quite same as of SMGF and PMGF except few

changes. Before elaborating the MMGF detection policy, some of the basic notations and terms that have been used in different phases of this scheme are summarized Table 7.

Example 5: Like we have done for previous two gate faults, here also we explain the *MMGF* localization policy using an example and then we summarize all the steps in Algorithm3.

Step 1: Again consider the same circuit of Figure 5(a) and now in the very first step, we compute the input-output combinations (RS_{FF}) for the circuit C_{FF}. The computed values are displayed in Figure 7(a).

Step 2: By following the test vector selection criteria for PMGF, we find the following four patterns {$\langle 0, 0, 1\rangle$, $\langle 0, 1, 1\rangle$, $\langle 1, 0, 1\rangle$, $\langle 1, 1, 1\rangle$ for circuit C_{FF} that can detect all the possible *MMGFs* occurring due to missing of gates $G_0$and G_1. So, $T_{G_0G_1} = \langle 1, 3, 5, 7\rangle$.

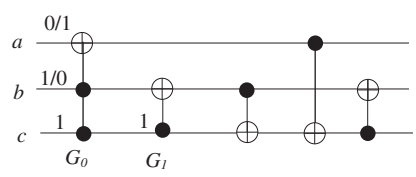
In the similar way, other test sets like $T_{G_1G_2} = \langle 1, 2, 3, 5, 6\rangle$, $T_{G_2G_3} = \langle 1, 2, 3, 4\rangle$, $T_{G_3G_4} = \langle 2, 3, 4, 5, 6, 7\rangle$ are obtained.

Table 7: Notations and their interpretations for Algorithm 3

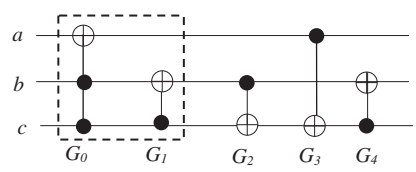
Notations	Definition
$T_{G_iG_j}$	Test set for detection of possible missing gates G_i and G_j from the circuit.
$U_{G_iG_j}$	Unique test set for the detection of possible missing gates G_i and G_j from the circuit.

$$RS_{FF} = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 2 & 1 & 4 & 7 & 5 & 6 & 3 \end{pmatrix}$$

(a)



(b)



(c)

Figure 7: (a) Input-output combinations for C_{FF} of Figure 5(a) (b). Test vector generation for G_0G_1 of C_{FF} (c) Testable Circuit (C_{TEST}) for MMGF

Table 8: Occurrence Factor Table for *MMGF*

Test Values in T	Occurrence Factor (O_J)
1	3
2	3
3	4
4	2
5	3
6	2
7	2

Step 3: Now, we compute the test set T for the entire circuit.

$$T = T_{G_0G_1} \cup T_{G_1G_2} \cup T_{G_2G_3} \cup T_{G_3G_4}$$

$$T = \langle 1, 3, 5, 7\rangle \cup \langle 1, 2, 3, 5, 6\rangle \langle 1, 2, 3, 4\rangle$$

$$\cup \langle 2, 3, 4, 5, 6, 7\rangle$$

$$T = \langle 1, 2, 3, 4, 5, 6, 7\rangle.$$

Step 4: Next, to compute the occurrence factor of $O_1$for $J = 1$, we search through the values present in $T_{G_0G_1}$, $T_{G_1G_2}$, $T_{G_2G_3}$, $T_{G_3G_4}$ and find that 1 is present in first three sets – $T_{G_0G_1}$, $T_{G_1G_2}$, $T_{G_2G_3}$, so, the value for O_1 becomes 3. Similarly, we compute rest of the occurrence factors and find that $O_2 = 3$, $O_3 = 4$, $O_4 = 2$, $O_5 = 3$, $O_6 = 2$ and $O_7 = 2$.

Step 5: Once the computation is over, now, we form a list containing occurrence factors and their respective test values. The newly form occurrence list is given in Table 8.

Step 6: Now, we need to form $U_{G_0G_1}$ and while computing this parameter we find that $m = O_7 = 2$, so $U_{G_0G_1} = \langle 1, 7\rangle$. Similarly, we find rest of the values and they are: $U_{G_1G_2} = \langle 1, 6\rangle$, $U_{G_2G_3} = \langle 1, 4\rangle$ and $U_{G_3G_4} = \langle 6, 7\rangle$.

Step 7: In this step, the test set T_L for the entire circuit is computed by concatenating all the unique test set values.

$$T_L = U_{G_0G_1} \cup U_{G_1G_2} \cup U_{G_2G_3} \cup U_{G_3G_4}$$

$$T_L = \langle 1, 7\rangle \cup \langle 1, 6\rangle \cup \langle 1, 4\rangle \cup \langle 6, 7\rangle$$

$$T_L = \langle 1, 4, 6, 7\rangle$$

Step 8: Now, in this step we will detect and localize the *MMGF* type fault if present in the circuit with 100% accuracy. Let's consider that gate G_0 and G_1 are missing (shown with dotted box in Figure 7(c)) from the circuit and thereby an *MMGF* fault has occurred in C_{TEST}.

Now, we randomly choose a value for (K) from the set $T_L = \langle 1, 4, 6, 7\rangle$ and let it be 7. We find that $RS_{FF}(7) \neq P_{TEST}(7)$, so, it is confirmed that there is an *MMGF* type fault in C_{TEST}, but the fault yet not has localized.

ALGORITHM 3: Detection and Localization of MMGF in Reversible Circuit

Input: A testable reversible circuit (C_{TEST}).
Output: Exact level where fault is present or name of the gates in which fault is present.
begin

Step1: **Computation of Reversible Specification (RS_{FF}):**
Reversible specification (RS_{FF}) of the fault free circuit (C_{FF}) is computed.

Step2: **Test Vector ($T_{G_jG_k}$) Generation for Each *MMGF* due to Missing of Gates(G_jG_k):**
(2^n-1) possible inputs are applied to the C_{FF} and test set ($T_{G_jG_k}$) for each possible missing of gates (G_jG_k) of C_{FF} is generated by setting '1' at all the control nodes of either G_j or G_k and either '0'or '1' to target node and through line of gates, where $0 \le j \le (N-1)$ and $0 \le k \le (N-1)$.

Step3: **Complete Test Set (T) for All Possible *MMGF*:**
Complete test set (T) of the circuit as follows:
$T = T_{G_0G_1} \cup T_{G_1G_2} \cup T_{G_2G_3} \cup \ldots \cup T_{G_{N-2}G_{N-1}}$.

Step4: **Computation of Occurrence Factor (O_J):**
Occurrence factor (O_J) of each vector (J) in T is calculated, where $J \in T$.

Step5: **Formation of Occurrence Factor Table:**
The occurrence factor table stores occurrence of different vector of T.

Step6: **Computation of Unique Test Set ($U_{G_{j,k}}$) for Each Gate:**
Compute unique test vector ($U_{G_jG_k}$)for each possible missing gates G_j and G_k by combining lowest occurrence test vector with another equal or higher occurrence test vector in ($T_{G_jG_k}$).

Step7: **Computation of Test Set for Fault Localization (T_L):**
Test set (T_L) of the circuit is computed as follows:
$T_L = U_{G_0G_1} \cup U_{G_1G_2} \cup U_{G_2G_3} \cup \ldots \cup U_{G_{N-2}G_{N-1}}$.

Step8: ***MMGF* Detection and Exact Localization:**
Apply any test vectors (K) randomly from $T_L$on the C_{TEST} and store the output in $P_{TEST}(K)$, where $K \in T_L$.
IF ($RS_{FF}(K) = P_{TEST}(K)$) THEN
begin

8.1 *MMGF* fault is present in C_{TEST}.
//Detection of MMGF in C_{TEST}//

8.2 Find out $T_{G_jG_k}$.such that, K $\in T_{G_jG_k}$and corresponding $U_{G_jG_k}$ of each $T_{G_jG_k}$would be usedto localize the fault.
//Exact Localization of *MMGF* in C_{TEST}//

In way to do that we check the sets which contain 7 and find it is $T_{G_0G_1}$ and $T_{G_3G_4}$. So, the fault may have occurred due to missing of either G_0G_1 or G_3G_4 gates. But once we check the circuit for $K = 1$, we find that $RS_{FF}(1) \neq P_{TEST}(1)$ which now has confirmed that the fault has originated due to missing of gate G_0 and G_1.

5. TIME COMPLEXITY OF OUR TESTING ALGORITHM

Here, we analyze the time complexity of our testing algorithm.

Each of our testing process (for example Algorithm1) contains eight steps and the cumulative complexity of those phases is the computational complexity of our test algorithm.

So, the cumulative complexity (T)

$$
\begin{aligned}
&= complexityof(step1 + step2 + step3 + step4 \\
&\quad + step5 + step6 + step7 + step8) \\
&= O(2^n) + O(2^n) + O(N) + O(N.n) + O(N) \\
&\quad + O(N.n) + O(N) + O(2^n) \\
&= O(3.2^n) + O(3.N) + O(2N.n) \\
&\approx O(2^n)
\end{aligned}
$$

where n is the number of input lines and N is the number of gates present in a circuit. Hence, $T \approx O(2^n)$.

From the obtained equation it is very evident that the computational time is primarily dependent on n and the gates present in the circuit (N) merely contribute to the incurred complexity.

But, if we consider only the fault localization phase of the algorithm, then the computational time of that particular phase lies between $O(1)$ to $O(2^n)$ and this is based on the selection of a random value from the test set T_L.

In the best case, the computational time of the algorithm is $O(1)$ and this is only possible if the fault is identified at the very first attempt.

6. EXPERIMENTAL RESULTS

In order to verify the correctness of our fault detection model, we have tested it over a wide pool of benchmarks [28] which include Arithmetic, non-arithmetic, and Symmetric functions. Apart from this type of benchmarks, we have generated some miscellaneous functions (synthetic benchmarks) and executed them over the testing model. The obtained results are summarized in six result tables (from Tables 9–14), where the first three tables contain test results from small size functions and the last three tables summarize the experimental evaluation from medium and large size benchmarks suites.

The parameters that are considered in our result tables are, Input specification for the benchmarks (C_{FF}), total

Table 9: Results for SMGF on small Benchmark functions

Fault Free Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *SMGF* in the C_{FF}	Unique Test Vector for Exact Localization of Fault (U_{G_i})	Cross Check Required by the Vectors	Value Of I For SMGF	Test size Reduced By (%)
Ham3tc	3	5	(1,2,3,4,5,6,7)	(1,2,3,4,6,7)	G_0	(3,7)	(1)	3	15%
					G_1	(1,3)	–	2	
					G_2	(1,2)	–	2	
					G_3	(6,4)	–	2	
					G_4	(2,4)	–	2	
Peres Gate	3	2	(1,3,5,7)	(1,3)	G_0	(3)	(1)	2	50%
					G_1	(1)	–	1	
N^{th} prime	3	4	(1,2,3,4,5,6,7)	(1,2,3,5,6)	G_0	(6)	–	1	28%
					G_1	(3,5)	(1)	3	
					G_2	(1,3)	–	2	
					G_3	(2)	–	1	
Ham3	3	5	(1,2,3,4,5,6,7)	(1,2,3,4,6,7)	G_0	(6,7)	(4)	3	15%
					G_1	(4,6)	–	2	
					G_2	(2,3)	–	2	
					G_3	(1,3)	–	2	
					G_4	(1,2)	–	2	
Rd32d1	4	4	(4,5,6,7,8,9,10,11,12,13,14,15)	(4,6,7,10,12,13,14)	G_0	(12,13)	(10)	3	47%
					G_1	(10,14)	–	2	
					G_2	(6,7)	(4)	3	
					G_3	(4)	–	1	
317_tc	3	6	(0,1,2,3,4,5,6,7)	(0,1,2,3,4,5,6,7)	G_0	(1)	–	1	0%
					G_1	(4,7)	(1)	3	
					G_2	(2,7)	(1)	3	
					G_3	(0,5)	(1,3)	4	
					G_4	(0,6)	(1,3)	4	
					G_5	(3,5)	(1)	3	
Xor5d1	5	4	(2,3,4,5,6,7,8,9, 10,11,12,13,14,15, 16,17,18,19,20,21, 22,23,24,25,26, 27,28,29,30,31)	(2,6,12, 24)	G_0	(24)	–	1	87%
					G_1	(12)	–	1	
					G_2	(6)	–	1	
					G_3	(2)	–	1	

(*continued*)

Table 9: Continued.

Fault Free Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *SMGF* in the C_{FF}	Unique Test Vector for Exact Localization of Fault (U_{G_i})	Cross Check Required by the Vectors	Value Of l For SMGF	Test size Reduced By (%)
Mod5d1	5	8	(2,3,4,5,6,7,8,9, 10,11,12,13,14, 15,16,17,18,19, 20,21,22,23,24, 25,26,27,28, 29,30,31)	(2,4,6,8,12,16,18,24)	G_0	(18)	(2,16)	3	74%
					G_1	(6)	(2,4)	3	
					G_2	(2)	–	1	
					G_3	(12)	(4,8)	3	
					G_4	(4)	–	1	
					G_5	(24)	(8,16)	3	
					G_6	(8)	–	1	
					G_7	(16)	–	1	
Mod5d2	5	9	(0,1,2,3,4,5, 6,7,8,9,10,11,12, 13,14,15,16,17, 18,19,20,21,22, 23,24,25,26,27, 28,29,30,31)	(0,4,8,10,17,21,28,30)	G_0	(0,28)*	–	2	74%
					G_1	(0,28)*	–	2	
					G_2	(10)	(0,28)	3	
					G_3	(30)	(0,4,28)	4	
					G_4	(4,28)	(0)	3	
					G_5	(21)	(0,4,28)	4	
					G_6	(17)	(0,28)	3	
					G_7	(0)	(8,17,28)	4	
					G_8	(8)	(0,28)	3	
Hwb4d3	4	11	(1,2,3,4,5,6,7,8, 9,10,11,12,13,14,15)	(1,3,4,5,6,7,8, 10,11,12,13,14)	G_0	(1,5)	–	2	20%
					G_1	(11,14)	(4,5,6,8,12)	7	
					G_2	(1,10)	–	2	
					G_3	(1,3)	(4,5,6,8,11,12,14)	9	
					G_4	(4,8)	–	2	
					G_5	(2,4,7,11)	(4,8)	6	
					G_6	(6,8)	–	2	
					G_7	(3,13,14)	–	3	
					G_8	(12,14)	(5,6)	4	
					G_9	(5,6)	–	2	
					G_{10}	(10,13)	–	2	

*Exact localization is not possible for G_0 and G_1 in circuit *mod5d2*.

Table 10: Results on PMGF over small Benchmark circuits

Fault Free Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *PMGF* in Various Gates of C_{FF} (G_i, x)	Unique Test Vector for Exact Localization of Fault (U_{G_i}, x)	Cross Check Required by the Vectors	Value of I For PMGF	Test size Reduced By (%)
Ham3tc	3	5	(0,1,2,3,4,5,6,7)	(1,2,3,4,5,6,7)	(G_0, b)	(5)	-	1	14%
					(G_0, c)	(2,6)	(4)	3	
					(G_1, c)	(4,6)	-	2	
					(G_2, b)	(3,4)	-	2	
					(G_3, a)	(1,7)	-	2	
					(G_4, c)	(3,6)	-	2	
Peres Gate	3	2	(0,1,2,4,6,7)	(0,1,2)	(G_0, b)	(1)	-	1	50%
					(G_0, c)	(2)	(0)	2	
					(G_1, c)	(0)	-	1	
N^{th} prime	3	4	(0,1,2,3,4,5,6,7)	(0,1,2,4,5,6,7)	(G_0, a)	(0,1)	-	2	14%
					(G_1, b)	(7)	-	1	
					(G_1, c)	(2,4)	(6)	3	
					(G_2, c)	(4,6)	-	2	
					(G_3, b)	(5)	-	1	
Ham3	3	5	(0,1,2,3,4,5,6,7)	(1,2,3,5,6,7)	(G_0, a)	(2,3)	(1)	3	25%
					(G_0, b)	(5)	-	1	
					(G_1, a)	(1,2)	-	2	
					(G_2, b)	(1,6)	-	2	
					(G_3, c)	(2,7)	-	2	
					(G_4, a)	(3,6)	-	2	
Rd32d1	4	4	(0,1,2,3,4,5, 6,7,8,9,10, 11,12,13,14,15)	(0,2,4,6,8,10,13,14)	(G_0, a)	(4,6)	-	2	47%
					(G_0, b)	(10)	-	1	
					(G_1, a)	(0,4)	-	2	
					(G_2, b)	(2,14)	13	3	
					(G_2, c)	(4,8)	-	2	
					(G_3, b)	(13)	-	1	
317_tc	3	6	(0,1,2,3,4,5,6,7)	(1,2,3,4,5,6,7)	(G_1, a)	(1)	-	1	15%
					(G_2, c)	(4,6)	-	2	
					(G_3, b)	(2,7)	(1)	3	
					(G_3, c)	(3,6)	(4)	3	
					(G_4, a)	(5)	-	1	
					(G_4, b)	(4,7)	(1)	3	
					(G_5, b)	(1,7)	-	2	
Xor5d1	5	4	(0,1,2,3,4,5,6, 7,8,9,10,11,12, 13,14,15,18,19, 20,21,22,23,24, 25,26,27,28,29, 30,31)	(5,8,18,30)	(G_0, a)	(8)	-	1	87%
					(G_1, b)	(5)	(8)	2	
					(G_2, c)	(18)	-	1	
					(G_3, d)	(30)	-	1	

(*continued*)

Table 10: Continued.

Fault Free Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *PMGF* in Various Gates of C_{FF} (G_i, x)	Unique Test Vector for Exact Localization of Fault (U_{G_i}, x)	Cross Check Required by the Vectors	Value of I For PMGF	Test size Reduced By (%)
Mod5d1	5	8	(0,1,2,3,4,5, 6,7,8,9,10,11,12, 13,14,15,16,17, 18,19,20,21,22,23, 24,25,26,27,28,29)	(0,2,4,7,14,15,16,18, 22,23,26,27,28,29)	(G_0, a)	(2,14)	(0)	3	40%
					(G_0, d)	(16,28)	(0)	3	
					(G_1, c)	(18,19)	(26)	3	
					(G_1, d)	(4,28)	(0)	3	
					(G_2, d)	(0,28,29)	-	3	
					(G_3, b)	(7,23)	(18)	3	
					(G_3, c)	(26,27)	(18)	3	
					(G_4, c)	(26,27)	-	2	
					(G_5, a)	(14,15)	(0)	3	
					(G_5, b)	(18,22)	(23)	3	
					(G_6, b)	(18,22,23)	-	3	
					(G_7, a)	(0,14,15)	-	3	
Mod5d2	5	9	(0,1,2,3,4,5,6, 7,8,9,10,11,12, 13,14,15,16,17,18, 19,20,21,22,23,24, 25,26,27,28,29,30,31)	(0,3,5,7,8,11, 13,15,21,22, 24,26,29,30)	(G_2, a)	(22,26,30)	(21)	4	44%
					(G_2, d)	(0,5,8)	-	3	
					(G_3, c)	(3,11)	(0)	3	
					(G_3, d)	(5,13,21)	-	3	
					(G_4, c)	(0,3,24)	-	3	
					(G_5, b)	(7)	-	1	
					(G_5, c)	(0,3,29)	-	3	
					(G_6, b)	(15,24,26)	-	3	
					(G_7, a)	(5,21,22)	-	3	
					(G_7, b)	(8,15,22)	-	3	
					(G_8, a)	(21,24,29)	-	3	
Hwb4d3	4	11	(0,1,2,3,4,5, 6,7,8,9,10, 11,12,13,14,15)	(1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15)	(G_0, d)	(2,4)	-	2	14%
					(G_1, b)	(2,5)	-	2	
					(G_2, c)	(3,11)	-	3	
					(G_3, b)	(7)	(2)	2	
					(G_3, d)	(4,6)	(2)	3	
					(G_4, a)	(9)	-	1	
					(G_5, a)	(10,12,15)	(9)	4	
					(G_5, c)	(1,8)	-	2	
					(G_6, b)	(1,10)	-	2	
					(G_7, b)	(1,4)	(6)	3	
					(G_7, d)	(6,7)	-	2	
					(G_8, a)	(10,13)	-	2	
					(G_9, d)	(4,7)	-	2	
					(G_{10}, c)	(1,3)	-	2	

Table 11: Results for MMGF on small Benchmark circuits

Original Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *MMGF* in Various Gates of C_{FF}	Unique Test Vector for Exact Localization of Fault (U_{G_i})	Cross Check Required by the Vectors	Value of I for MMGF	Test size Reduced By (%)
Ham3tc	3	5	(1,2,3,4,5,6,7)	(1,4,6,7)	G_0 G_1	(1,7)	-	2	43%
					G_1 G_2	(1,6)	-	2	
					G_2 G_3	(1,4)	-	2	
					G_3 G_4	(6,7)	-	2	
Peres Gate	3	2	(1,3,5,7)	(1)	G_0 G_1	(1)	-	1	75%
N^{th} *prime*	3	4	(1,2,3,4,5,6,7)	(1,2,5,6)	G_0 G_1	(6)	-	1	43%
					G_1 G_2	(1,5)	-	2	
					G_2 G_3	(2)	-	1	
Ham3	3	5	(1,2,3,4,5,6,7)	(1,3,4,5)	G_0 G_1	(4,5)	(3)	3	43%
					G_1 G_2	(3,4)	-	2	
					G_2 G_3	(1,4)	-	2	
					G_3 G_4	(1,3)	-	2	
Rd32d1	4	4	(4,5,6,7,8,9,10, 11,12,13,14,15)	(4,6,12,13)	G_0 G_1	(13)	(6)	3	67%
					G_1 G_2	(6,12)	-	2	
					G_2 G_3	(4)	-	1	
317_tc	3	6	(0,1,2,3,4,5,6,7)	(1,3,4,5,6,7)	G_0 G_1	(1)	-	1	25%
					G_1 G_2	(4)	-	1	
					G_2 G_3	(7)	(4)	2	
					G_3 G_4	(5,6)	(4)	3	
					G_4 G_5	(3,5)	-	2	
Xor5d1	5	4	(2,3,4,5,6,7,8, 9,10,11,12,13,14, 15,16,17,18,19, 20,21,22,23,24, 25,26,27,28, 29,30,31)	(2,20,24)	G_0 G_1	(24)	-	1	90%
					G_1 G_2	(12)	-	1	
					G_2 G_3	(6)	-	1	
Mod5d1	5	8	(2,3,4,5,6, 7,8,9,10,11,12, 13,14,15,16,17, 18,19,20,21,22, 23,24,25,26,27, 28,29,30,31)	(2,5,6,7,8, 16,18,24,26)	G_0 G_1	(7,18,26)	-	3	69%
					G_1 G_2	(2,26)	-	2	
					G_2 G_3	(2,6)	-	2	
					G_3 G_4	(5)	(24)	2	
					G_4 G_5	(24)	-	1	
					G_5 G_6	(8)	(16)	2	
					G_6 G_7	(16)	-	1	

(*continued*)

Table 11: Continued.

Original Circuit (C_{FF})	Number of Input lines (n)	Number of Gates (N)	Total Set of Test Vector for the Circuit (T)	Set of Test Vectors to Localize the Fault (T_L)	Possible *MMGF* in Various Gates of C_{FF}	Unique Test Vector for Exact Localization of Fault (U_{G_i})	Cross Check Required by the Vectors	Value of I for MMGF	Test size Reduced By (%)
Mod5d2	5	9	(0,1,2,3,4,5, 6,7,8,9,10,11, 12,13,14,15,16, 17,18,19,20,21, 22,23,24,25,26, 27,28,29,30,31)	(8,16,17,24, 6,27,28,29)	G_0 G_1	(24,27)*	-	-	75%
					G_1 G_2	(24,27)*	-	-	
					G_2 G_3	(26)	(24,27)	3	
					G_3 G_4	(28,29)	(24,27)	4	
					G_4 G_5	(28,29)	(24,27)	4	
					G_5 G_6	(16,17)	(24,27)	4	
					G_6 G_7	(8)	(24,27)	3	
					G_7 G_8	(8)	(24,27)	3	
Hwb4d3	4	11	(1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15)	(1,2,5,6,7,8, 10,11,14,15)	G_0 G_1	(1,5,7)	-	3	29%
					G_1 G_2	(1,6,10)	-	3	
					G_2 G_3	(1,4,10,,15)	(6)	5	
					G_3 G_4	(1,8,12,14)	-	4	
					G_4 G_5	(1,8,14)	(12)	4	
					G_5 G_6	(2,8,11)	(1)	4	
					G_6 G_7	(2,8)	(1,11)	4	
					G_7 G_8	(3,4)	(1,6,8,15)	5	
					G_8 G_9	(2,5,15)	(5,6)	4	
					G_9 G_{10}	(5,7)	(1)	3	

*Exact localization is not possible for G_0G_1 and G_1G_2 in the circuit *mod5d2*.

Table 12: Results for SMGF on medium and large Benchmark functions

Original Circuit (C_{FF})	Number Of Input Lines (n)	Number Of Gates (N)	Possible SMGF in C_{FF}	Total Test Vector Needed For SMGF Detection (T)	Total Test Vector Needed For Exact Localization (T_L)	Test Size Reduced by (%)
mod5adders	6	15	15	60	31	49%
rd53d1	7	12	12	126	115	9%
rd53d2	8	12	12	248	193	23%
random_001	9	4	4	374	296	42%
rd73_140	10	20	20	1011	774	24%
rd73_141	10	14	14	935	564	40%
random_002	10	10	10	921	493	47%
random_003	10	20	20	1020	817	20%
random_004	11	10	10	1993	1736	13%
symd9_146	‘12	28	28	3905	3381	19%
random_005	13	19	19	7889	7013	13%
random_006	14	22	22	16005	11878	26%
rd84d1	15	28	28	32546	31303	5%
random_007	15	20	20	32324	28879	12%
random_008	16	25	25	65411	58302	13%
random_009	17	20	20	130029	109081	17%
random_010	18	21	21	261737	211532	20%
random_011	19	26	26	524011	453902	15%
random_012	20	25	25	1048423	948423	14%
cm150a_210	21	53	53	2095152	1894853	11%

Table 13: Results for PMGF from medium and large Benchmark circuits

Original Circuit (C_{FF})	Number Of Input Lines (n)	Number Of Gates (N)	Possible PMGF in C_{FF}	Total Test Vector Needed For PMGF Detection (T)	Total Test Vector Needed For Exact Localization (T_L)	Test Size Reduced by (%)
mod5adders	6	15	22	61	28	55%
rd53d1	7	12	28	119	107	12%
rd53d2	8	12	20	240	185	24%
random_001	9	4	20	397	305	25%
rd73_140	10	20	34	1021	694	33%
rd73_141	10	14	34	947	526	46%
random_002	10	10	140	910	485	47%
random_003	10	20	20	1015	701	31%
random_004	11	10	10	2031	1853	10%
symd9_146	‘12	28	49	4023	3567	12%
random_005	13	19	19	8005	6975	13%
random_006	14	22	22	16103	12365	24%
rd84d1	15	28	48	32312	31405	5%
random_007	15	20	20	32095	28043	14%
random_008	16	25	25	65510	59042	11%
random_009	17	20	20	129045	109123	16%
random_010	18	21	47	261894	211683	20%
random_011	19	26	59	523025	453816	15%
random_012	20	25	73	1049321	994100	14%
cm150a_210	21	53	205	2097398	1894782	8%

test vectors (T) used for the corresponding circuit, set of test vectors applied to the testable circuit (C_{TEST}) for fault localization (T_L), the possible areas of faults in C_{FF} circuits, total number of test vectors required, the number of iterations (I) required for localizing fault and the percentage reduction in the size of the total test set (T).

Though our approach performs fairly, but it too has some limitations that are summarized here.

i. As our approach has exponential computational complexity ($O(2^n)$), we had to restrict the testing process up-to 21 input variables.

ii. We cannot exactly specify the number of iterations (I) required to test a circuit but we have seen that it varies in-between $(X-Y) \leq I \leq (X+Y)$, where $X = T_L/2$, $Y = T_L/3$ and T_L is total number of test vectors required to localize the fault in the circuit.

iii. In our testing scheme we only have shown the first-order PMGF and two gates missing for MMGF to detect and localize the faults, but this approach can be generalized to detect higher-order PMGF and MMGF beyond 2-gates.

So, all the limitations that we have mentioned above can be investigated further to improve the testing scheme.

Table 14: Results for MMGF over medium and large Benchmark circuits

Original Circuit (C_{FF})	Number Of Input Lines (n)	Number Of Gates (N)	Possible MMGF in C_{FF}	Total Test Vector Needed For MMGF Detection (T)	Total Test Vector Needed For Exact Localization (T_L)	Test Size Reduced by (%)
mod5adders	6	15	14	49	18	64%
rd53d1	7	12	11	97	85	13%
rd53d2	8	12	11	213	152	29%
random_001	9	4	3	303	217	28%
rd73_140	10	20	19	1013	780	24%
rd73_141	10	14	13	901	523	42%
random_002	10	10	9	865	476	45%
random_003	10	20	19	1025	789	23%
random_004	11	10	9	1873	1574	15%
sym9_146	‘12	28	27	3756	3119	17%
random_005	13	19	18	7765	6897	11%
random_006	14	22	21	15705	11267	28%
rd84d1	15	28	27	31546	30198	4%
random_007	15	20	19	30324	27965	8%
random_008	16	25	24	61411	53163	12%
random_009	17	20	19	126078	106081	15%
random_010	18	21	20	241737	201786	17%
random_011	19	26	25	523216	421549	20%
random_012	20	25	24	1044789	931436	11%
cm150a_210	21	53	52	2083587	1794513	13%

7. CONCLUSION

Several test generation algorithms are already been reported for fault model detection in reversible circuits. But effective and prompt diagnosis of the circuit can only be possible if fault localization can be performed. In this work, we have developed an algorithm that not only detects the fault but also identifies its location in the circuit. This has been established by executing our algorithm over various benchmark circuits and confirms both fault detection and localization using reduced test sets. Additionally, test set sizes are also reduced up to 87% for the *SMGF* and *PMGF* whereas 90% for the *MMGF*.

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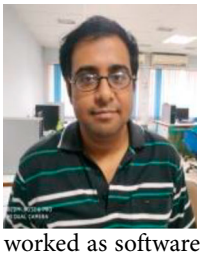

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